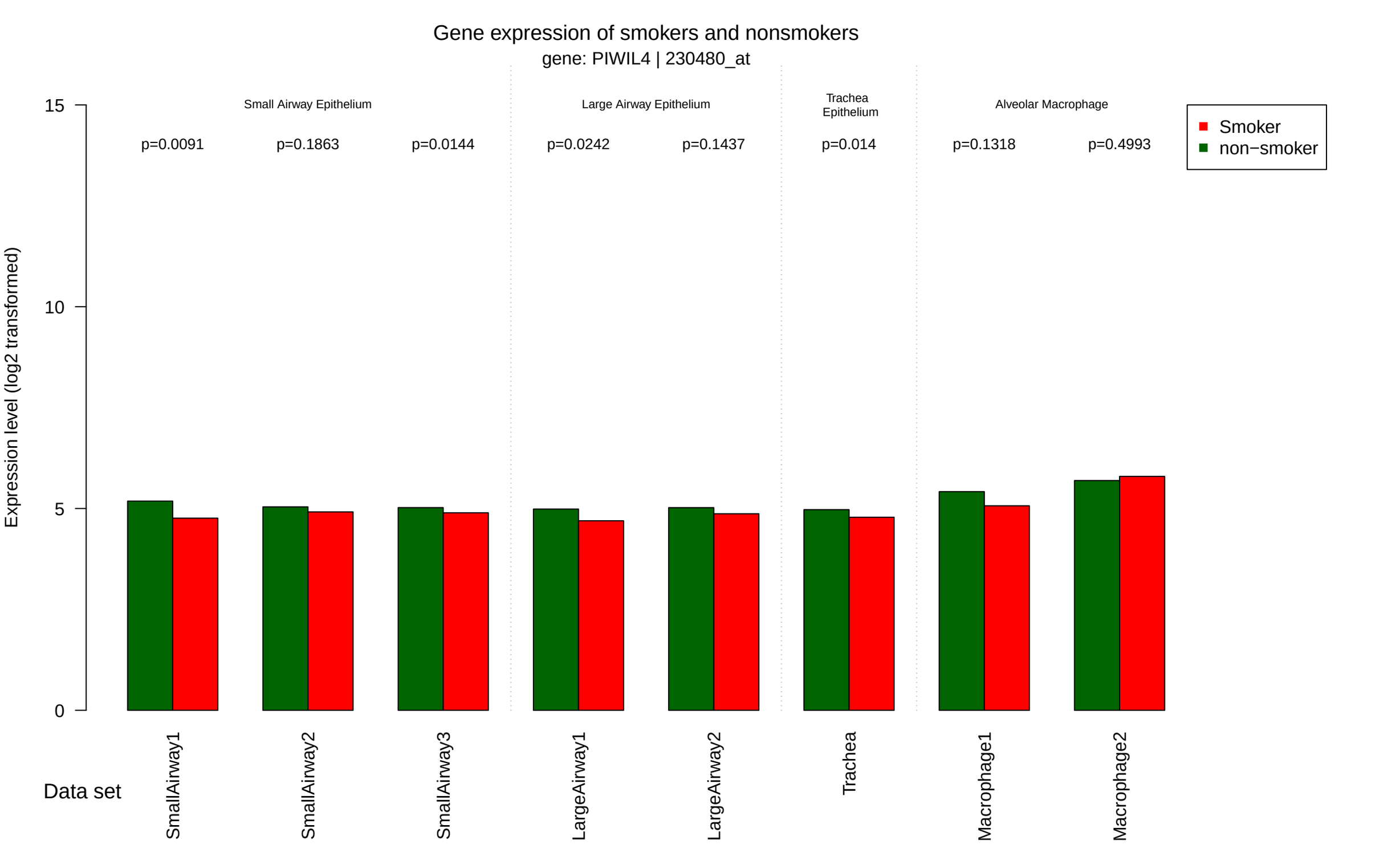
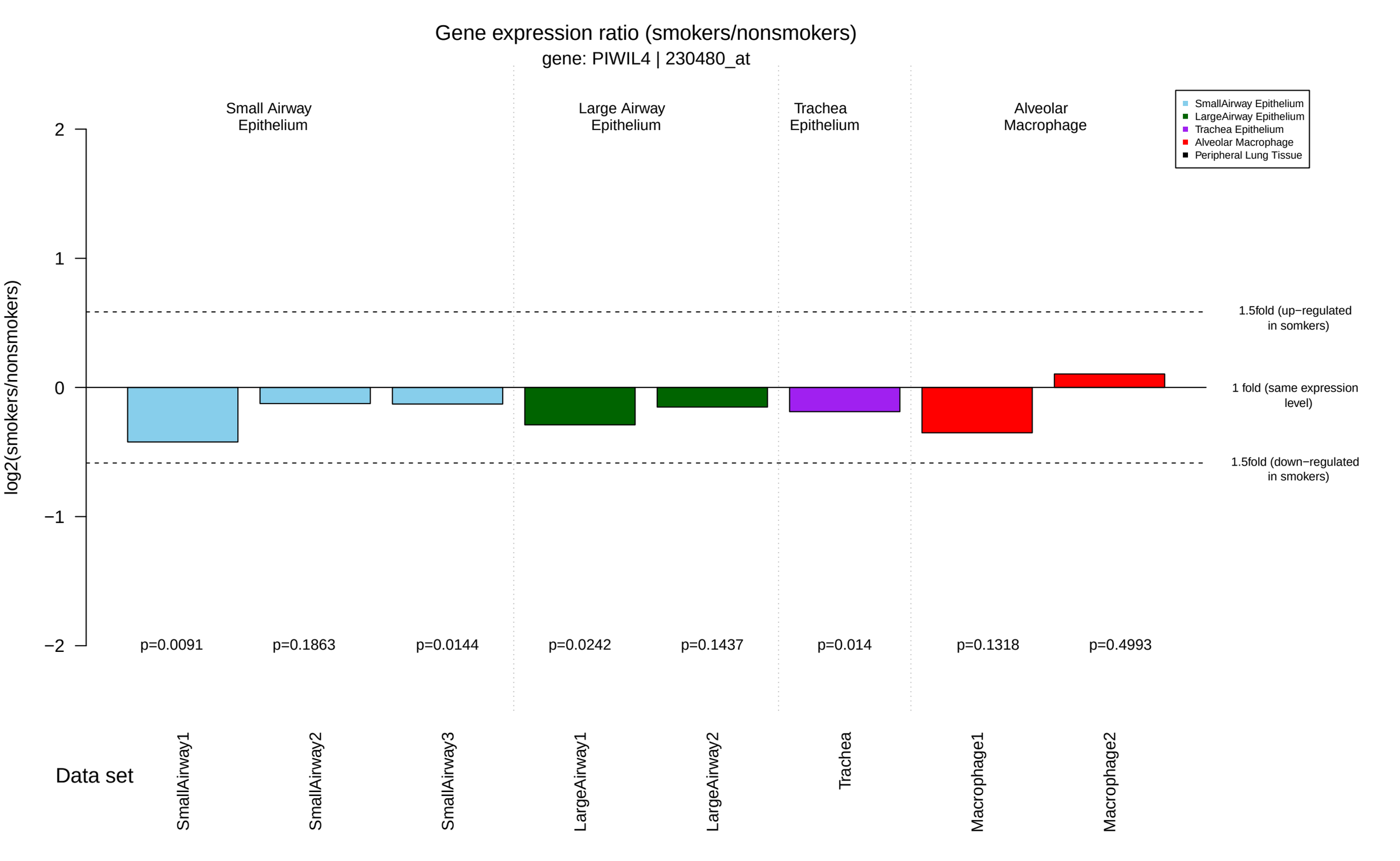
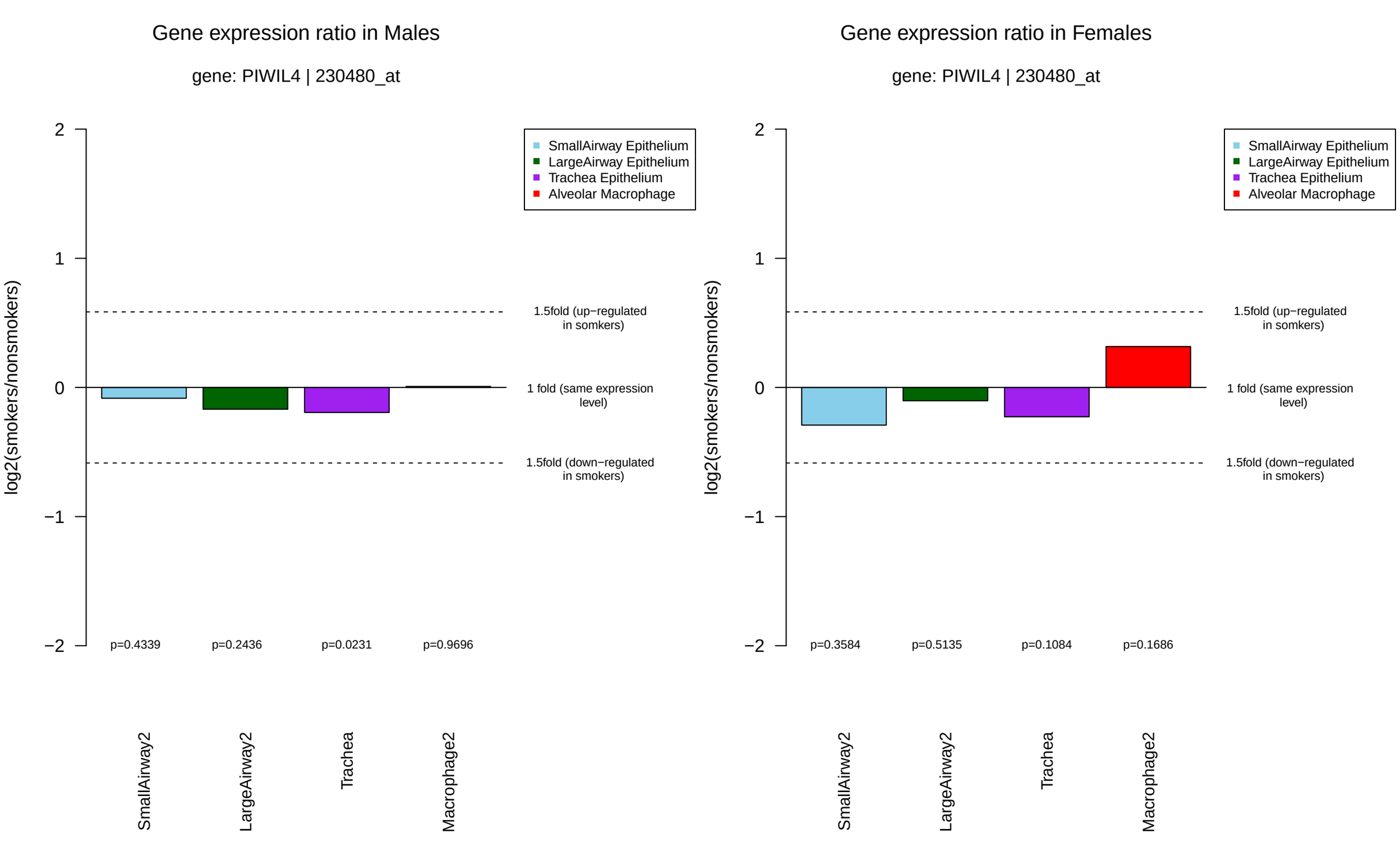
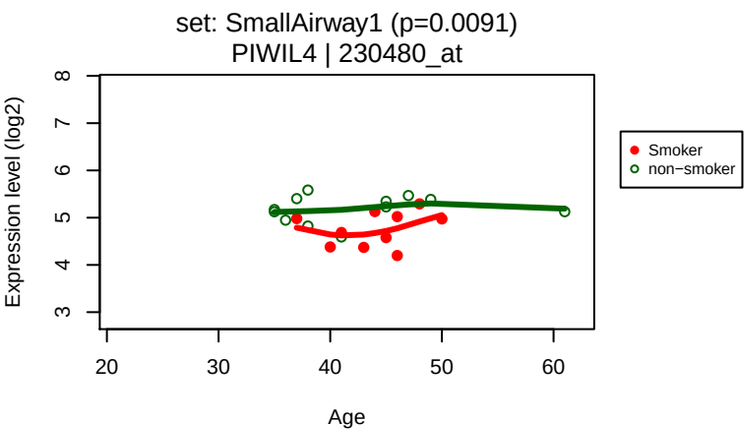
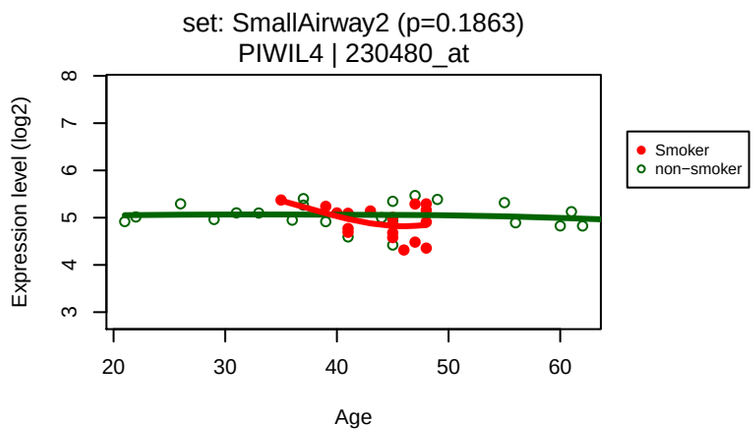
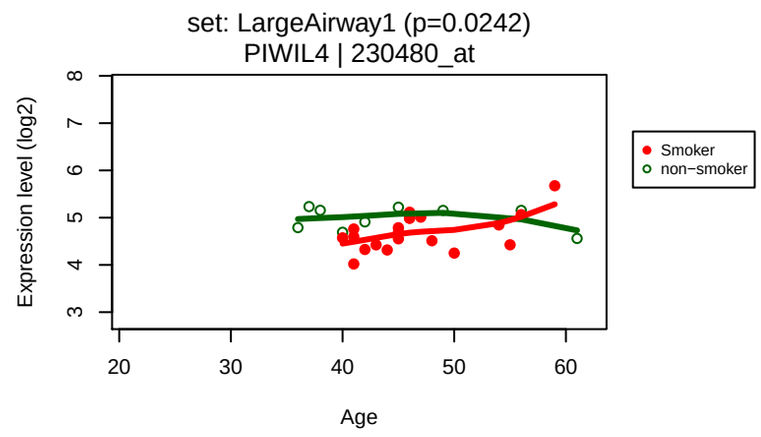

Gene expression of smokers and nonsmokers
gene: PIWIL4 | 230480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0091
p=0.1863
p=0.0144
p=0.0242
p=0.1437
p=0.014
p=0.1318
p=0.4993
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

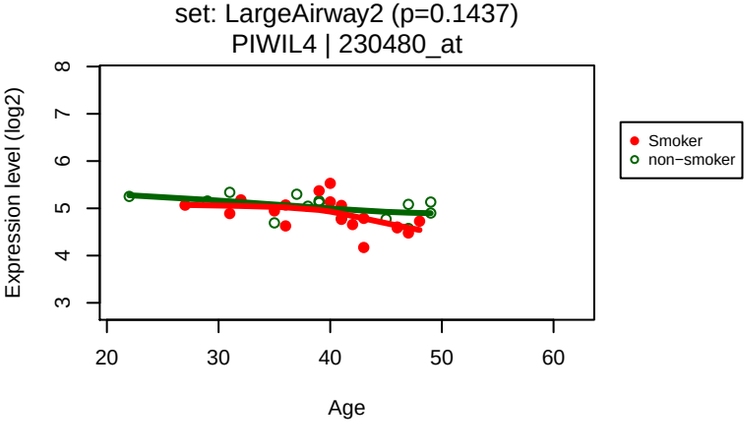

Gene expression ratio (smokers/nonsmokers)
gene: PIWIL4 | 230480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0091
p=0.1863
p=0.0144
p=0.0242
p=0.1437
p=0.014
p=0.1318
p=0.4993
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

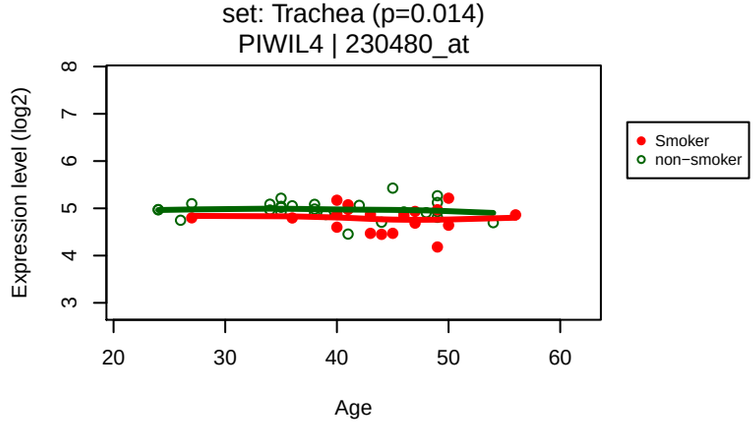

Gene expression ratio in Males
gene: PIWIL4 | 230480_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4339
p=0.2436
p=0.0231
p=0.9696
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIWIL4 | 230480_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3584
p=0.5135
p=0.1084
p=0.1686
SmallAirway2
LargeAirway2
Trachea
Macrophage2

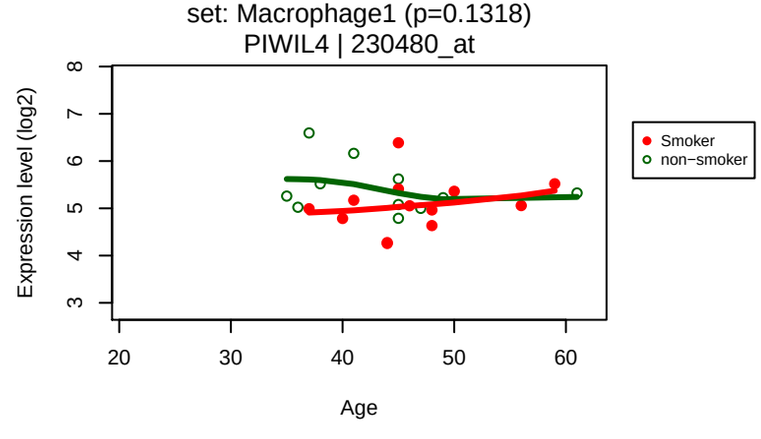

set: SmallAirway1 (p=0.0091)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker

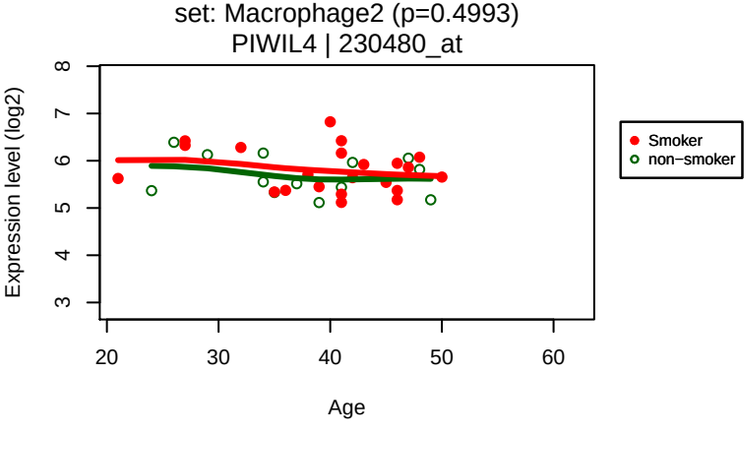

set: SmallAirway2 (p=0.1863)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker

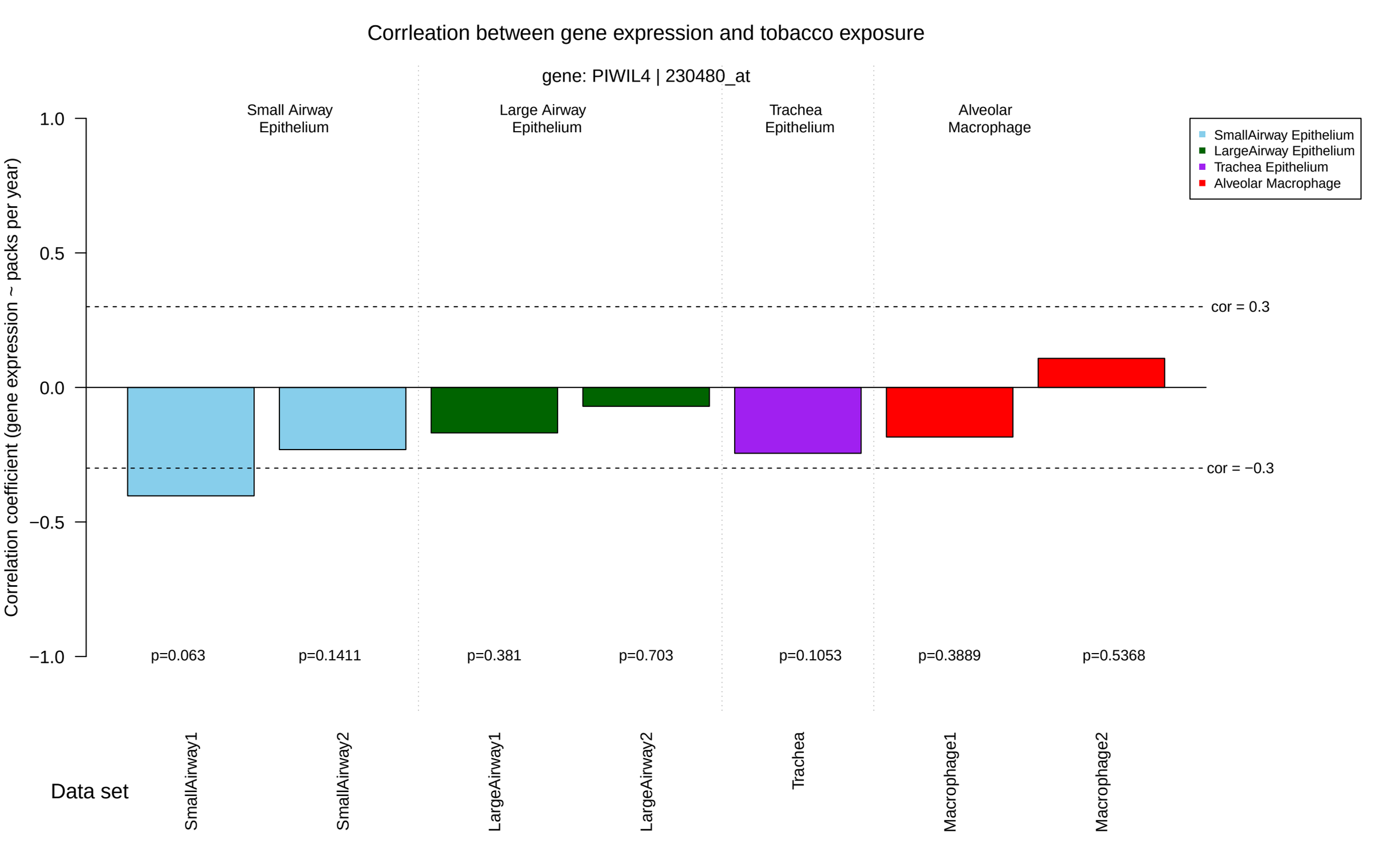

set: LargeAirway1 (p=0.0242)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1437)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.014)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1318)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4993)
PIWIL4 | 230480_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIWIL4 | 230480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.063
p=0.1411
p=0.381
p=0.703
p=0.1053
p=0.3889
p=0.5368
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2